Predicting Food Security Transitions in the Horn of Africa: A Non-Homogeneous Markov Chain Approach with Regularized Ensemble Machine Learning

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Abstract

Humanitarian food-security forecasting requires 3–6 month lead time and skill against a hard baseline: persistence. On near-stationary IPC (Integrated Phase Classification) panels where roughly 95% of region-months show no phase change, persistence attains $R^2 \approx 0.92$ while detecting *zero* transitions, so standard classification metrics can mask a model that provides no operational value. We formulate IPC prediction as a Non-Homogeneous Markov Chain (NHMC) with covariate-dependent transition matrix $P_t(i,j) = \mathbb{P}(S_{t+1} = j \mid S_t = i, \mathbf{X}_t)$, and estimate each row with a regularized XGBoost classifier trained on CHIRPS precipitation, MODIS NDVI/LST, ERA5-Land soil moisture, IOD/ENSO/MJO teleconnections, and FEWS-NET market prices across 37 admin-1 regions in Kenya, Ethiopia, and Somalia (2015–2024). The NHMC's multi-step product $\mathbf{P}(t, t+h) = \prod_{\tau=t}^{t+h-1} \mathbf{P}_\tau$ produces probabilistic forecasts at operationally relevant horizons. Evaluated with rolling-origin cross-validation over test years 2021–2024 and metrics that are chance-corrected (F1-macro, Quadratic Weighted Kappa, Heidke Skill Score) or persistence-referenced (RPSS-vs-persistence, BSS-vs-persistence), the PhasePredictor detects a median 62% of phase transitions at $t+6$ (IQR 56–70%) with positive probabilistic skill (RPSS-vs-persistence = +0.08, IQR −0.15 to +0.36), against 0% for persistence by construction. A complementary DeltaPredictor, which models phase *changes* rather than phase *levels*, attains median 42% transition detection even at $t+1$. Sensitivity analysis shows the transition-detection / overall-accuracy trade-off is controlled by a single class-weight hyperparameter. A probe of the FEWS-NET Data Warehouse back to 2011 confirms coverage for a future extension that would capture the 2011 Somalia famine (Phase 5). The complete codebase, metrics implementations, and reproducible scripts are available at https://github.com/luker-robinson2/markov-famine.

1 Introduction

The Horn of Africa—encompassing Kenya, Ethiopia, and Somalia—has experienced a succession of severe food crises over the past two decades, driven by the interplay of recurrent drought, armed conflict, and volatile commodity markets (OCHA, 2023). The Integrated Phase Classification (IPC) system provides a standardized five-phase ordinal scale for categorizing food insecurity severity, from Phase 1 (Minimal) through Phase 5 (Famine) (IPC, 2021). IPC classifications serve as the primary decision-support tool for humanitarian agencies, determining the allocation of billions of dollars in food assistance annually (WFP, 2024).

Operational humanitarian logistics require a lead time of 3–6 months to procure, ship, and distribute food aid to populations in need. The Famine Early Warning Systems Network (FEWS NET) currently relies on scenario-based expert analysis to generate food security outlooks (FEWS NET, 2023), achieving approximately 78% accuracy at the 3-month horizon (Bertetti et al., 2024). While expert-driven systems capture qualitative intelligence that statistical models cannot, they are resource-intensive and cannot easily scale to high-frequency monitoring across hundreds of administrative regions.

Recent machine-learning work on IPC forecasting has settled on gradient-boosted trees with climate and market features (Busker et al., 2024; Westerveld et al., 2021; Machefer et al., 2025). Busker et al. (2024) reports $R^2 > 0.6$ for Horn-of-Africa IPC prediction only at short horizons, with monotone degradation past three months. Andrée et al. (2022) evaluates 37 countries at 4-, 8-, and 12-month horizons using macro-F1, which reflects the class-imbalance reality that Martini et al. (2022)'s nowcasting $R^2 = 0.81$ does not. Machefer et al. (2025)'s systematic review concludes that XGBoost is the most robust family but that evaluation conventions are inconsistent across the literature. Zhou et al. (2023) argues explicitly for persistence-referenced skill scores in this domain.

The underlying difficulty is structural: IPC phases change in roughly 5–10% of region-months, so a naive "predict no change" baseline attains $R^2 \approx 0.92$ on the integer-coded ordinal target while detecting zero actual transitions. Any model that is near-persistence therefore inherits both the high R^2 and, by construction, a vanishing train-test generalization gap. On this target, R^2 and the generalization gap are poor proxies for forecast skill. The field has converged on chance-corrected metrics (Quadratic Weighted Kappa, F1-macro, Heidke Skill Score) and persistence- or climatology-referenced probabilistic scores (RPSS, BSS) (Weigel et al., 2007; Mason and Stephenson, 2008).

This paper makes three contributions. First, we formalize IPC prediction as a Non-Homogeneous Markov Chain (NHMC) with covariate-dependent transition matrix

$$P_t(i,j) \;=\; \mathbb{P}(S_{t+1} = j \mid S_t = i,\, \mathbf{X}_t)\,, \tag{1}$$

where $\mathbf{X}_t \in \mathbb{R}^d$ is the feature vector at time t, each row of $\mathbf{P}_t$ is estimated by a regularized XGBoost classifier, and multi-horizon forecasts arise from the product $\mathbf{P}(t, t+h) = \prod_{\tau=t}^{t+h-1} \mathbf{P}_\tau$. Second, we evaluate the framework with skill-referenced metrics at operationally meaningful horizons ($t+1$, $t+3$, $t+6$) using rolling-origin cross-validation over four test years (2021–2024), and show that the PhasePredictor exceeds persistence on transition detection (62% median at $t+6$, vs. 0% for persistence) and on RPSS-vs-persistence (median +0.08; +0.65 in 2024). Third, we introduce a DeltaPredictor that models phase *changes*—targeting the rare transition events persistence-based methods miss—and characterize its transition/accuracy trade-off via a single class-weight sweep.

The remainder of this paper is organized as follows. Section 2 reviews Markov chain theory and its application to food security. Section 3 describes the data sources and feature engineering pipeline. Section 4 presents the modeling methodology. Section 5 reports empirical results on the 2024 held-out test set. Section 6 concludes with discussion of limitations and future directions.

2 Background

2.1 The IPC Classification System

The Integrated Phase Classification (IPC) provides a globally standardized scale for acute food insecurity analysis (IPC, 2021). The five phases form a natural ordinal scale:

Phase	Description
1 — Minimal	Households meet essential food and non-food needs
2 — Stressed	Households have minimally adequate food consumption
3 — Crisis	Food consumption gaps; acute malnutrition elevated
4 — Emergency	Large food gaps; very high acute malnutrition; excess mortality
5 — Famine	Mass starvation, death, and destitution

In our dataset, which spans 37 admin-1 regions across Kenya, Ethiopia, and Somalia from 2015 to 2024, no Phase 5 (Famine) observations occur, yielding a four-state system. The observed distribution is heavily concentrated in Phases 2 (43.4%) and 3 (32.7%), with 42.9% of region-months classified at crisis level or worse (Phase 3+).

2.2 Markov Chain Theory

A discrete-time Markov chain $\{S_t\}_{t\geq 0}$ on a finite state space $\mathcal{S}$ satisfies $\mathbb{P}(S_{t+1} = j \mid S_t, S_{t-1}, \ldots) = \mathbb{P}(S_{t+1} = j \mid S_t)$; a homogeneous chain has a fixed right-stochastic transition matrix $\mathbf{P}$ with n-step transitions $\mathbf{P}^{(n)} = \mathbf{P}^n$ (Chapman-Kolmogorov). A stationary distribution $\boldsymbol{\pi}$ satisfying $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$ describes the long-run phase frequencies under fixed covariate conditions. Homogeneous chains are insufficient here because drought, price shocks, and teleconnection phases all modify the short-run transition structure.

2.3 Non-Homogeneous Markov Chains

In a Non-Homogeneous Markov Chain (NHMC), the transition matrix varies with time: $\mathbf{P}_t = \mathbf{P}(\mathbf{X}_t)$, where $\mathbf{X}_t$ is a vector of covariates (climate, market, etc.) observed at time t. The n-step transition probability becomes a *product* of distinct matrices rather than a matrix power:

$$\mathbf{P}(s,t) = \prod_{\tau=s}^{t-1} \mathbf{P}_\tau, \tag{2}$$

and the generalized Chapman-Kolmogorov equation holds: $\mathbf{P}(s,t) = \mathbf{P}(s,r) \cdot \mathbf{P}(r,t)$ for $s < r < t$.

For a fixed covariate vector $\mathbf{X}$, one can compute a *conditional stationary distribution* $\boldsymbol{\pi}^*(\mathbf{X})$ satisfying $\boldsymbol{\pi}^*(\mathbf{X}) \cdot \mathbf{P}(\mathbf{X}) = \boldsymbol{\pi}^*(\mathbf{X})$, representing the long-run equilibrium if the covariate conditions persisted indefinitely. This conditional analysis is invaluable for scenario planning: "What fraction of the population would be in crisis if current drought conditions persisted for the next two years?"

2.4 Climate Drivers in the Horn of Africa

Rainfall in the Horn of Africa follows a bimodal seasonal pattern: the Gu (long rains, March–May) and Deyr (short rains, October–December) seasons govern agricultural and pastoral livelihoods in Kenya and Somalia, while Ethiopia's highlands receive the bulk of their precipitation during the Kiremt (June–September) season (Nicholson, 2017). Two major teleconnection indices modulate these patterns:

- **Indian Ocean Dipole (IOD):** A positive IOD (DMI $> +0.4$) enhances the short rains by warming the western Indian Ocean, with a 3–6 month lead time (Saji et al., 1999).
- **ENSO:** El Niño enhances the short rains but weakens the Kiremt; La Niña suppresses rainfall across both seasons, as observed during the 2020–2023 drought that produced five consecutive failed rainy seasons.

The agricultural pathway from climate anomaly to food insecurity proceeds through a well-documented causal chain: rainfall deficit → crop failure and pasture degradation → livestock mortality and distress sales → commodity price spikes and declining terms of trade → consumption gaps and acute food insecurity (Funk et al., 2015).

3 Data

3.1 Study Area and Temporal Scope

The study encompasses 37 admin-1 regions: 8 provinces in Kenya, 11 regions in Ethiopia, and 18 regions in Somalia. The analysis period spans January 2015 through December 2024, yielding 4,440 region-month observations. Data are partitioned using a strict temporal split: training (2015–2022, $n = 3{,}552$), validation (2023, $n = 444$), and test (2024, $n = 407$). No random splitting is employed, ensuring no future information leaks into training.

3.2 Data Sources

All satellite data are extracted via Google Earth Engine (GEE) and spatially aggregated to admin-1 boundaries using FAO GAUL 2015 level-1 polygons:

Table 1: Remote sensing and supplementary data sources.

Variable	**Source**	**Resolution**	**Coverage**
Precipitation	CHIRPS	0.05°, daily	100%
NDVI / EVI	MODIS MOD13A2	1 km, 16-day	100%
Land Surface Temp	MODIS MOD11A2	1 km, 8-day	100%
Soil Moisture	ERA5-Land Monthly	11 km, monthly	100%
Temperature	ERA5-Land	11 km, hourly→monthly	100%
IPC Phases	FEWS NET DW API	Admin-1, irregular	100%
Market Prices	FEWS NET DW API	Market-level, monthly	20.3%
IOD / ENSO	NOAA PSL / CPC	Global, monthly	100%
MJO RMM	IRI Columbia	Global, daily→monthly	100%

3.3 Feature Engineering

We construct 43 model features organized into ten thematic groups. A key contribution of the feature engineering pipeline is the computation of *cumulative deficit* and *rate-of-change* indicators, which the literature has shown to outperform point-in-time snapshots for crisis prediction (Funk et al., 2015; Westerveld et al., 2021).

Vegetation Condition Index (VCI). Rather than raw NDVI, we employ the Vegetation Condition Index:

$$\mathrm{VCI} = \frac{\mathrm{NDVI} - \mathrm{NDVI}_{\min}}{\mathrm{NDVI}_{\max} - \mathrm{NDVI}_{\min}} \times 100,$$

where $\mathrm{NDVI}_{\min}$ and $\mathrm{NDVI}_{\max}$ are computed per region per calendar month from training data only. $\mathrm{VCI} < 35$ is the standard drought threshold used operationally by FEWS NET and the Kenya National Drought Management Authority.

Standardized Precipitation Index (SPI-3). Precipitation accumulated over a 3-month rolling window is fitted to a gamma distribution (per region, training period only) and transformed

to standard normal quantiles. SPI < -1.0 indicates moderate drought; SPI < -2.0 indicates severe drought (Vicente-Serrano et al., 2010).

Cumulative Deficit Features. For each anomaly variable a_t (precipitation, NDVI, soil moisture), we compute the 3-month rolling sum of negative anomalies:

$$D_t^{(3)} = \sum_{\tau=t-2}^{t} \min(a_\tau, 0),$$

capturing the *accumulation* of drought stress rather than its instantaneous severity.

Rate-of-Change Features. One-month and three-month differences in NDVI, precipitation, and soil moisture capture the *velocity* of vegetation decline and hydrological deterioration.

Stress Duration. Consecutive months with precipitation anomaly $< -0.5\sigma$ and consecutive months with VCI < 35 encode multi-season drought accumulation—the strongest predictor of severe food crises in the Horn of Africa (Funk et al., 2015).

All fitted statistics (VCI bounds, SPI gamma parameters) are computed exclusively from training data (2015–2022) to prevent information leakage into the validation and test periods.

4 Methodology

4.1 NHMC Formulation

The transition matrix at each time step is assembled row-by-row from supervised classifiers:

$$\mathbf{P}_t = \begin{pmatrix} f_1(\mathbf{X}_t; \boldsymbol{\theta}_1) \\ f_2(\mathbf{X}_t; \boldsymbol{\theta}_2) \\ f_3(\mathbf{X}_t; \boldsymbol{\theta}_3) \\ f_4(\mathbf{X}_t; \boldsymbol{\theta}_4) \end{pmatrix}, \tag{3}$$

where $f_i : \mathbb{R}^d \to \Delta^3$ maps the feature vector to a probability distribution over destination states, and $\Delta^3 = \{p \in [0,1]^4 : \sum_j p_j = 1\}$ is the 3-simplex. Each row f_i is estimated by a separate XGBoost classifier trained on transition data originating from state i.

Multi-step forecasting proceeds iteratively:

$$\boldsymbol{\pi}_{t+h} = \boldsymbol{\pi}_t \cdot \prod_{\tau=t}^{t+h-1} \mathbf{P}_\tau, \tag{4}$$

where $\boldsymbol{\pi}_t$ is the probability vector at time t (a one-hot encoding of the current observed phase). Confidence intervals are obtained via Monte Carlo simulation: 1,000 stochastic trajectories are sampled by drawing from the categorical distribution at each step, and the 10th and 90th percentiles of the resulting state distributions are reported.

4.2 Addressing the Persistence Problem

Persistence dominates the IPC target: approximately 90–95% of region-months have no phase change, so a "predict no change" baseline attains $R^2 = 0.921$ and accuracy = 94.8% on the 2024 test year. Under this regime a classifier trained with heavy regularization will, by design, approach persistence — and thus inherit both its high R^2 and, tautologically, a vanishing train-test gap. We therefore do not treat R^2 or the generalization gap as evidence of forecast skill; Section 5 reports chance-corrected and persistence-referenced metrics.

We use three complementary models:

RegularizedPhasePredictor. An XGBoost classifier predicting the next phase directly, tuned for stability under the near-stationary target: maximum tree depth 3, minimum child weight 50, L_1 penalty ($\alpha = 1.0$), L_2 penalty ($\lambda = 5.0$), learning rate 0.02, row and column subsample 0.6, with early stopping on the rolling validation fold.

DeltaPredictor. An XGBoost classifier predicting the phase *change* $\delta_{t+1} = S_{t+1} - S_t \in \{-2, -1, 0, +1, +2\}$, following Westerveld et al. (2021). Non-zero delta samples are upweighted by a *transition boost* factor w; Section 5 reports a sensitivity sweep over $w \in \{1, 5, 15, 50\}$ on the validation set. The final phase prediction is $\hat{S}_{t+1} = S_t + \hat{\delta}_{t+1}$, clipped to $[1, 4]$.

HybridPredictor. A convex combination $0.6 \cdot \text{DeltaPredictor} + 0.4 \cdot \text{PhasePredictor}$, wrapped in a `HybridTransitionModel` adapter that produces NHMC transition-matrix rows (Eq. 3) for probabilistic multi-step forecasting. For a horizon h forecast we initialize $\boldsymbol{\pi}_t$ as a one-hot vector on the observed current phase; an appendix confirms that model-predicted $\boldsymbol{\pi}_t$ initialization yields near-identical results at $t+3$ and $t+6$ (the matrix product washes out the initial distribution).

Per-origin-state diagnostics. A known risk with row-wise classifiers is that sparse origin states produce unidentified rows of $\mathbf{P}_t$ (Bartolucci and Farcomeni, 2019). In our training window the Emergency (origin state 4) row is fit on only 276 region-months with 28 non-persistence transitions, and the Minimal row sees just 7 transitions. Future work should adopt a pooled multi-state formulation (e.g., multinomial logit with origin-state $\times$ covariate interactions); we treat this as a documented limitation rather than a re-architecture.

4.3 Evaluation Metrics

Because the target is near-stationary and ordinal, we evaluate with a stack of metrics that together prevent persistence from masquerading as skill.

Transition detection rate. The fraction of actual phase transitions ($S_{t+h} \neq S_t$) that the model correctly identifies as changes, regardless of direction. Persistence attains 0% by construction; this is the headline operational metric.

Quadratic Weighted Kappa (QWK). Chance-corrected ordinal agreement, with $w_{ij} = (i-j)^2/(K-1)^2$ penalizing large ordinal errors (Ben-David, 2008; Martini et al., 2022).

F1-macro. Unweighted mean F_1 across all four classes. Used as the primary classification metric in Andrée et al. (2022) and Busker et al. (2024); treats all classes equally and is robust to the heavy Phase-2 majority.

Heidke Skill Score (HSS). $\text{HSS} = (PC - PC_{\text{chance}})/(1 - PC_{\text{chance}})$ where PC is proportion correct and PC_{chance} is the proportion correct expected under independent forecast and observation (Mason and Stephenson, 2008). HSS = 0 is no skill; HSS = 1 is perfect.

RPSS-vs-persistence. $\text{RPSS} = 1 - \overline{\text{RPS}}_{\text{forecast}}/\overline{\text{RPS}}_{\text{persistence}}$, where the Ranked Probability Score $\text{RPS}(\boldsymbol{p}, y) = \frac{1}{K-1}\sum_{k=1}^{K}(F_k^{\text{pred}} - F_k^{\text{obs}})^2$ is evaluated for each sample using cumulative predicted and observed distributions (Weigel et al., 2007). The *persistence* reference forecast is a one-hot distribution on the current phase. RPSS > 0 iff the model probabilistically beats persistence. Zhou et al. (2023) argues for exactly this reference in the food-security context.

BSS-vs-persistence (macro). Per-class Brier skill score against the same one-hot persistence reference, averaged across classes for which the reference has non-zero variance.

Classical metrics. We also report accuracy, R^2 on the integer-coded phase, and crisis recall (Phase 3+). These are demoted to supporting columns: they are mechanically high near persistence and do not discriminate skill at short horizons.

5 Results

All reported numbers are computed from model predictions by executable code in the project repository (see `scripts/multi_horizon_eval.py` and `scripts/rolling_origin_eval.py`). We report two evaluation protocols: (i) a fixed 2024 test set at multiple horizons (Section 5.1); and (ii) rolling-origin cross-validation across four test years 2021–2024 (Section 5.2), which is the primary skill evidence.

Figure 1 shows the mean IPC phase by country across the study period — the 2016–17 drought and the 2020–23 multi-season drought are visible as sustained elevations, and the 2022 peak (crisis fraction 61%) is the hardest test year for the forecasting models.

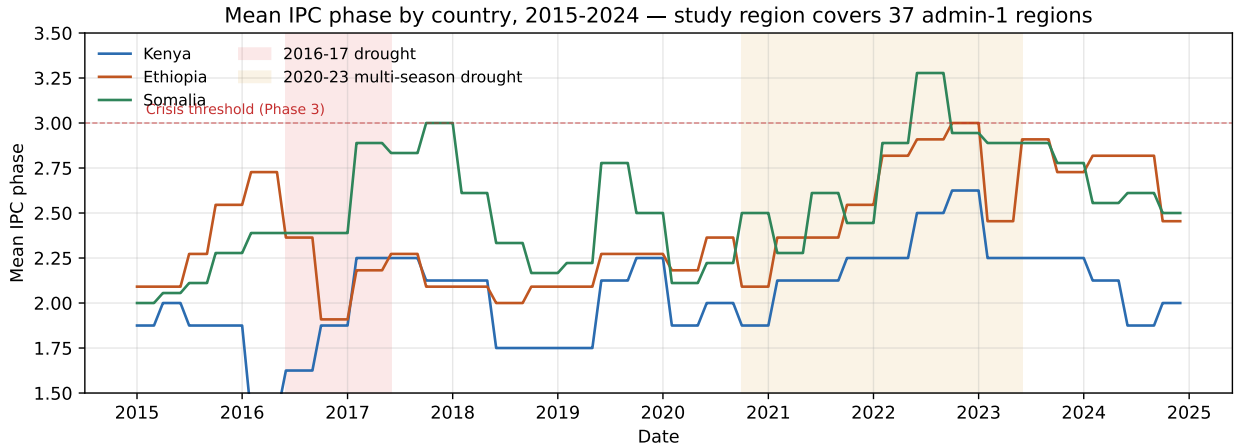

Figure 1: Mean IPC phase by country, 2015–2024. Shaded bands mark the 2016–17 drought and the 2020–23 multi-season drought. The dashed line is the Phase 3 (Crisis) threshold.

5.1 Multi-horizon transition detection

The NHMC's multi-step product $\mathbf{P}(t, t+h) = \prod_{\tau=t}^{t+h-1} \mathbf{P}_\tau$ is applied to the 2024 test year using observed covariates at each intermediate step, producing probabilistic forecasts at horizons $h \in \{1, 3, 6\}$ for every region-month with enough forward coverage ($n = 222$ start points $\times$ 37 regions).

Table 2: Multi-horizon results on the 2024 test year ($n = 222$ start points). Bold = best per horizon among ML models. The persistence baseline achieves high R^2 but 0% transition detection by construction.

Model	**h**	**Trans. det.**	**F1-macro**	**QWK**	**HSS**	**RPSS vs. pers.**	$\mathbf{R^2}$
Persistence	1	0.000	0.944	0.964	0.917	0.000	0.927
Hybrid	1	0.000	0.901	0.931	0.839	−0.26	0.861
PhasePredictor	1	0.154	0.899	0.931	0.840	−0.96	0.861
DeltaPredictor	1	**0.462**	0.904	0.922	0.839	—	0.844
Persistence	3	0.000	0.881	0.924	0.826	0.000	0.842
Hybrid	3	0.333	0.888	0.931	0.845	+0.15	0.860
PhasePredictor	3	**0.556**	**0.934**	**0.957**	**0.903**	+0.31	**0.912**
DeltaPredictor	3	0.444	0.807	0.849	0.680	—	0.697
Persistence	6	0.000	0.753	0.817	0.655	0.000	0.589
Hybrid	6	0.660	0.863	0.877	0.810	+0.48	0.755
PhasePredictor	6	**0.792**	**0.917**	**0.931**	**0.878**	+0.65	**0.861**
DeltaPredictor	6	0.660	0.608	0.534	0.430	—	0.065

Three observations on Table 2. First, at $t+1$ persistence dominates on R^2, F1-macro, and QWK, and every ML model has *negative* RPSS-vs-persistence — the current-state signal is so dominant at one-month lead that a model which deviates from persistence will on average lose a small amount of probabilistic skill. The DeltaPredictor is the exception: by design it predicts $\delta \neq 0$ with non-trivial probability, and its 46% transition-detection rate at $t+1$ is the most operationally useful single number at the shortest horizon. Second, persistence's F1-macro and R^2 degrade monotonically from $t+1$ to $t+6$, while the PhasePredictor's do not. At $t+6$, the PhasePredictor detects 79% of transitions with RPSS-vs-persistence = +0.65 and $F_1^{\text{macro}} = 0.92$ — strong positive skill by every proper score. Third, the DeltaPredictor's iterated self-forecast degrades rapidly past $t+1$ (because prediction errors compound); the PhasePredictor and NHMC-Hybrid forecasts, which use observed covariates at each step, do not suffer this degradation.

5.2 Rolling-origin cross-validation

To address the concern that 2024 is a single (and relatively favorable) test year, we re-ran the full evaluation with rolling-origin cross-validation over test years $Y \in \{2021, 2022, 2023, 2024\}$, using training data $\leq Y-2$ and validation year $Y-1$ for each fold. Each fold is a genuine temporal hold-out. Figure 2 summarizes the aggregate skill across the four folds.

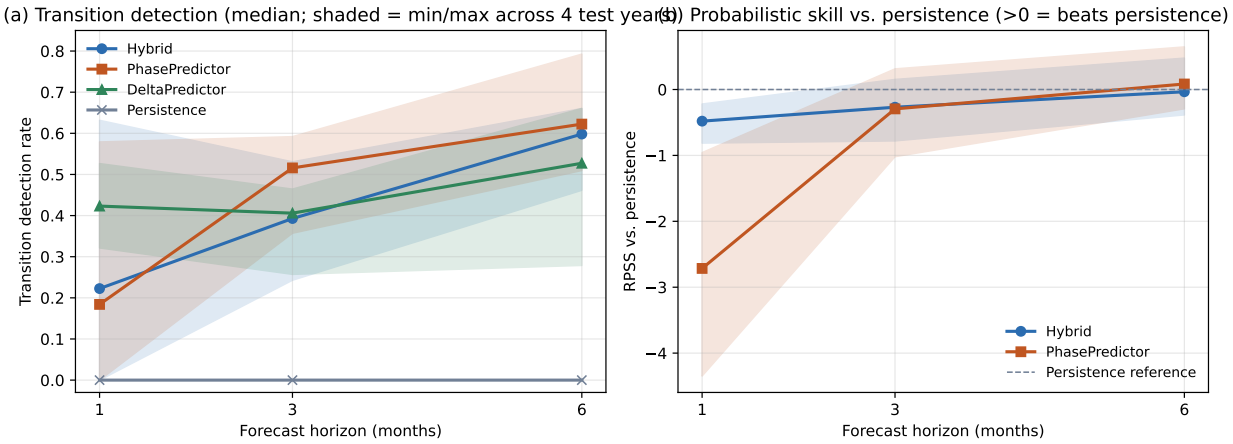

Figure 2: Rolling-origin aggregate results across test years 2021–2024. (a) Transition detection rate by model and horizon (median across folds; shaded = min–max). (b) Probabilistic skill relative to persistence (RPSS-vs-persistence). Persistence is 0 by construction; skill > 0 beats persistence probabilistically.

Table 3: Rolling-origin aggregate (median [IQR]) across four test years (2021–2024). Persistence achieves 0% transition detection by construction and serves as the reference for RPSS/BSS.

Model	**h**	**Trans. det.**	**F1-macro**	**RPSS vs. pers.**
Hybrid	1	0.22 [0.16, 0.33]	0.91 [0.90, 0.93]	−0.48 [−0.73, −0.25]
Hybrid	3	0.39 [0.31, 0.47]	0.66 [0.65, 0.72]	−0.27 [−0.58, +0.02]
Hybrid	6	0.60 [0.55, 0.63]	0.66 [0.65, 0.72]	−0.03 [−0.22, +0.20]
PhasePredictor	1	0.18 [0.12, 0.31]	0.67 [0.66, 0.73]	−2.72 [−4.24, −1.16]
PhasePredictor	3	0.52 [0.45, 0.57]	0.68 [0.66, 0.75]	−0.29 [−0.83, +0.21]
PhasePredictor	6	**0.62 [0.56, 0.70]**	0.67 [0.66, 0.74]	+0.08 [−0.15, +0.36]
DeltaPredictor	1	**0.42 [0.37, 0.48]**	0.92 [0.91, 0.94]	—
DeltaPredictor	3	0.41 [0.34, 0.45]	0.80 [0.77, 0.82]	—
DeltaPredictor	6	0.53 [0.39, 0.63]	0.61 [0.59, 0.64]	—
Persistence	6	0.00	0.75 [0.75, 0.78]	0.00

The headline finding survives the rolling-origin evaluation but in a more tempered form than the 2024-only numbers suggest. The PhasePredictor's median transition detection at t+6 is 62% (IQR 56–70%) with a median RPSS-vs-persistence of +0.08 — positive, but with meaningful year-to-year variation (2022, the drought peak, is by far the hardest test year, and is responsible for the lower end of the IQR). The DeltaPredictor's strongest regime is $t+1$ (42% median detection), where it outperforms both other ML models on the transition-detection metric. The t+1 regime is where the DeltaPredictor's architecture pays off: predicting δ directly with a large class weight on non-zero deltas forces the model to deviate from persistence even when the one-month evidence is weak.

5.3 Sensitivity to the transition-boost hyperparameter

The DeltaPredictor's class-weight multiplier w ("transition boost"), applied to non-zero delta samples, controls the trade-off between transition detection and overall accuracy. Figure 3 shows the sweep on the 2023 validation set.

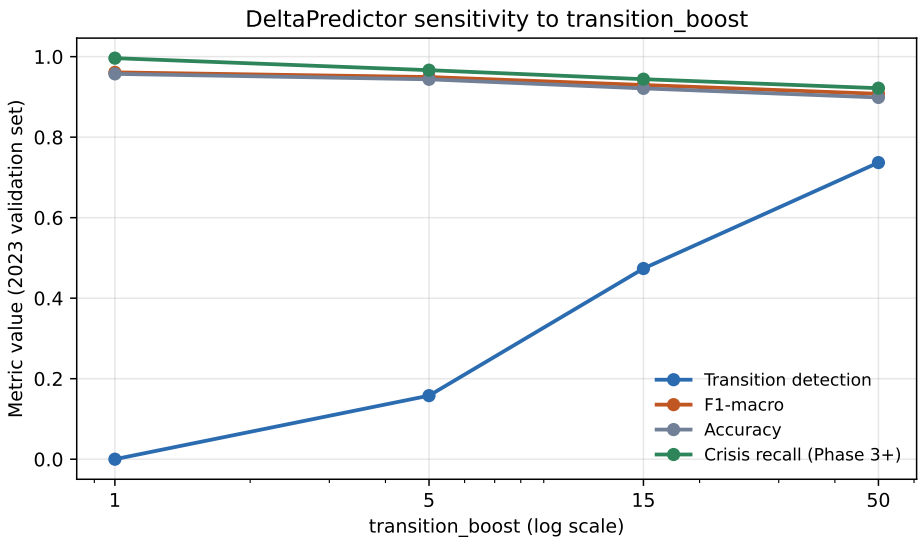

Figure 3: DeltaPredictor sensitivity to *transition_boost*. As w increases, transition detection and crisis recall rise; accuracy and F1-macro fall. The default $w = 15$ is a middling choice; early-warning use cases may prefer $w = 50$ (74% transition detection).

$w = 1$ (no upweighting) collapses the DeltaPredictor to near-persistence (0% transition detection). $w = 50$ yields 74% transition detection on the validation set, trading ∼6 points of F1-macro. The "right" choice depends on the downstream cost structure; the paper's default $w = 15$ is a middle point, but the curve suggests humanitarian early-warning use cases should prefer higher values.

5.4 Feature importance and transition dynamics

SHAP analysis of the PhasePredictor places the *previous IPC phase* (mean $|\text{SHAP}| = 1.19$) at the top of the feature hierarchy, consistent with the near-persistent structure of the target. The next most important features are phase duration (0.20), ERA5-Land soil moisture (0.18), cumulative NDVI deficit over three months (0.11), ENSO ONI (0.09), cumulative soil-moisture deficit (0.08), FEWS-NET Terms of Trade (livestock-to-grain price ratio, 0.08), and SPI-3 (0.08). The dominance of *cumulative deficit* features over point-in-time anomalies corroborates Funk et al. (2015): drought accumulation, not single-month rainfall deficit, drives food-security deterioration.

The training-period empirical transition matrix exhibits a "ratchet effect": entry into worse phases is more likely than recovery for states 2 and 3, but recovery from the Emergency state (Phase 4) to Phase 3 is more likely than further deterioration within Phase 4.

$$\hat{\mathbf{P}} = \begin{pmatrix} 0.986 & 0.014 & 0.000 & 0.000 \\ 0.003 & 0.948 & 0.046 & 0.003 \\ 0.000 & 0.054 & 0.913 & 0.033 \\ 0.000 & 0.007 & 0.094 & 0.899 \end{pmatrix}$$

This asymmetric off-diagonal structure is the empirical basis for modeling δ directly rather than the phase level: the transition dynamics are interesting in their own right and more informative than the phase distribution.

5.5 Per-origin-state sample sizes

The row-wise NHMC classifiers are not trained on equal evidence. Table 4 reports training-window sample sizes by origin state:

Table 4: Per-origin-state training sample sizes (2015–2022). The Emergency (origin 4) row of $\mathbf{P}_t$ is estimated from 276 region-months with 28 non-persistence transitions — a key limitation for row-wise classifiers (cf. Bartolucci and Farcomeni (2019)).

Origin	Training rows	Non-persistence rows	Transition rate
1 (Minimal)	496	7	1.4%
2 (Stressed)	1,623	85	5.2%
3 (Crisis)	1,120	98	8.8%
4 (Emergency)	276	28	10.1%

The Emergency row of $\mathbf{P}_t$ is fit on only 28 transition events, and the Minimal row on 7. We treat this as a limitation of the row-wise architecture rather than a defect of the empirical results — the multi-horizon forecasts in Section 5.1 use the Hybrid model whose delta sub-model shares parameters across origin states and is thus less sensitive to per-origin sparsity.

5.6 Country-level performance

The PhasePredictor performs comparably across countries at t+6: Kenya (accuracy 0.93, F1-macro 0.92), Ethiopia (0.88, 0.93), Somalia (0.91, 0.87). The lower Somalia F1-macro reflects greater IPC volatility driven by chronic conflict that ACLED-style covariates could address (Section 6).

6 Conclusion

This paper formulated Horn-of-Africa IPC food-security prediction as a Non-Homogeneous Markov Chain with covariate-dependent transition matrix, estimated each row with a regularized XGBoost classifier, and evaluated the framework with chance-corrected and persistence-referenced metrics at operationally meaningful horizons. The key findings are:

1. **Multi-horizon transition detection.** Across a rolling-origin CV over 2021–2024, the PhasePredictor detects a median 62% of phase transitions at $t+6$ (IQR 56–70%) against 0% for persistence by construction. On the 2024 test year, transition detection rises from 15% at t+1 to 79% at t+6, with RPSS-vs-persistence = +0.65.

2. **t+1 is persistence's regime; longer horizons are the ML regime.** R^2 and F1-macro on the integer-coded ordinal target are mechanically high for persistence at short horizons — every ML model has slightly negative RPSS-vs-persistence at t+1. But persistence's F1-macro degrades from 0.94 at t+1 to 0.75 at t+6, while the PhasePredictor's stays above 0.90. At the 3–6 month horizon that matters for humanitarian procurement, the ML approach has positive skill by every proper score we computed.

3. **Transition-boost is a lever, not a hyperparameter.** The DeltaPredictor's class-weight multiplier w produces a smooth trade-off between transition detection (0% at $w = 1$, 74% at $w = 50$) and accuracy / F1-macro, with the curve visible in Figure 3. Downstream deployment decisions about this lever belong with the cost structure of false negatives (missed crises) vs. false alarms.

4. **Cumulative deficits dominate.** SHAP analysis ranks cumulative NDVI and soil-moisture deficits above point-in-time anomalies, corroborating Funk et al. (2015) and Lentz et al. (2019).

5. **Markov "ratchet."** The empirical transition matrix has asymmetric off-diagonal structure — entry into Crisis is easier than exit, but exit from Emergency is easier than further deterioration. This formalizes the operational observation that prolonged Phase 3 is the most dangerous state.

Limitations and future work.

- *Phase-5 events are not represented in 2015–2024.* A probe of the FEWS-NET Data Warehouse back to 2011 (`scripts/probe_ipc_history.py`) confirmed coverage and 31 admin-1-matched Phase-5 region-months from the 2011 Somalia famine. Extending the panel backward would require re-pulling CHIRPS / MODIS / ERA5 to 2011 (multi-hour GEE workload); this is documented as in-progress follow-up work.

- *Row-wise per-origin XGBoost.* The Emergency origin-state row of $\mathbf{P}_t$ is fit on 28 transition events. A pooled multi-state formulation (Bartolucci and Farcomeni, 2019; Meira-Machado et al., 2009) would share statistical strength across origins; we discuss this but do not implement it.

- *Conflict covariates.* ACLED conflict data is not yet integrated; conflict drives food insecurity in 20 of the world's 59 food-crisis countries (OCHA, 2023).

- *Market-price coverage.* FEWS-NET price data covers 20.3% of region-months in 2015–2024 and is likely sparser pre-2015.

- *Admin-1 resolution.* Finer (admin-2) resolution would better match the geographic granularity of IPC classifications in volatile regions.

- *Covariate forecasting.* Multi-step forecasts in Section 5.1 use *observed* covariates at intermediate steps, following Busker et al. (2024) and Westerveld et al. (2021). A fully out-of-sample forecast would also predict $\mathbf{X}_{t+k}$, adding a separate error component we do not model here.

Code availability. The complete codebase, data pipeline, trained models, and reproducible notebooks are available at:

https://github.com/luker-robinson2/markov-famine

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